

Fake News Classification Using Semantic Text Analysis

Problem Statement

In today's digital era, the widespread dissemination of fake news on social media and online platforms poses a significant threat to public perception, trust, and decision-making. The goal of this project is to develop a reliable machine learning model to distinguish between fake and true news articles based on textual content using semantic classification techniques.

Methodology

1. Data Understanding & Preprocessing

- The dataset was split into training and validation sets.
- Preprocessing steps included:
 - Removal of special characters and punctuation.
 - Lowercasing (only when the subsequent character was uppercase to retain acronyms).
 - Lemmatisation focused on nouns.
 - Removal of stopwords.

2. Feature Engineering

- Word2Vec embeddings were used for semantic representation:
 - Loaded Google News pre-trained Word2Vec model (300-dimensional vectors).
 - Averaged word vectors for each article to represent the entire text.
 - Handled out-of-vocabulary and Nan tokens gracefully.

3. Model Building

- Applied multiple models:
 - Logistic Regression
 - Decision Tree
 - Random Forest
- Tuned hyperparameters using RandomizedSearchCV for efficient exploration.
- Evaluation metric prioritised: **F1 Score** (due to its balance between precision and recall).

Results

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.9095	0.9008	0.9105	0.9056
Random Forest	0.9094	0.9107	0.8981	0.9043
Decision Tree	0.8213	0.8308	0.7851	0.8073

- **Best Performing Model:** Logistic Regression
 - **Evaluation Metric Prioritized:** F1 Score (0.9056)
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Visualisations and Insights

- Word clouds showed fake news articles often use emotionally charged words.
 - True news had more neutral and factual terms.
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Inconsistencies and Outlier Handling

- Null or missing lemmatised strings were replaced with empty strings.
 - Articles with no valid vocabulary terms (from Word2Vec) were assigned zero vectors.
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Actionable Outcomes

- The model can be used by news aggregators and fact-checking organizations to flag potentially fake articles.
 - Integration with social media platforms can enable real-time detection of misinformation.
 - Can be extended to other languages or domains (e.g., health misinformation, financial scams).
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Conclusion

By leveraging semantic representations of text via Word2Vec and classifying with Logistic Regression, we built an effective and interpretable model for fake news detection. The model's strong performance in terms of F1 Score demonstrates its practical utility in addressing misinformation in real-world scenarios.

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Toolkits Used: Python, Scikit-learn, Gensim, NLTK, Matplotlib, Pandas